Enrida Ndreu

PhD Candidate in Quantitative Finance and Actuarial Data Science

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Profile

Mathematician by training, now completing a PhD at the University of Florence. My research applies stochastic analysis and machine learning to insurance and financial data, with a focus on *path signatures* (rough path theory), claims reserving, survival analysis, and risk measurement (VaR/ES) in a Solvency II setting. International research experience at TU Chemnitz and Universitat Pompeu Fabra.

Education

PhD in Social Sciences for Sustainability and Wellbeing 2023 – 2026 (expected)
Università degli Studi di Firenze Florence, Italy
39th cycle, funded doctoral scholarship. Research in quantitative risk modelling for insurance and finance.

Master's Degree in Finance and Risk Management 2020 – 2023
Università degli Studi di Firenze Florence, Italy
Final grade: **110/110 cum laude**.
Thesis: *Contingent Convertible Bonds: A New Asset Class. A Comparison between Valuation Models.*

Bachelor's Degree in Mathematics 2011 – 2014
Università degli Studi di Firenze Florence, Italy
Thesis: *Random Walks.*

Research

Interests: path signatures and rough paths; claims reserving and survival analysis; gradient boosting for claims severity; risk measures (VaR, Expected Shortfall); stochastic and spot volatility estimation.

Work in progress

- *A two-stage framework for insurance claims reserving.* Signature-based forecasting of aggregate paid flows for closed claims, combined with Cox proportional hazards, gradient-boosted severity models and VaR/ES for open-claim reserves.
- *Path signatures for financial volatility.* Augmenting signature-based models with Fourier estimates of spot volatility, with application to option-model calibration.

Research and Teaching Experience

Research Assistant Oct 2026 – Dec 2026
Università degli Studi di Firenze Florence, Italy
Signatures Volatility Models / Prof. Maria Elvira Mancino

Teaching Tutor – *Quantitative Finance and Derivatives and Computational Finance* Sep 2026 – Dec 2026
Università degli Studi di Firenze, MSc in Finance and Risk Management Florence, Italy

Visiting Researcher hosted by Prof. Elisa Alos Alcalde Feb 2026 – Apr 2026
Universitat Pompeu Fabra Barcelona, Spain

Visiting Researcher hosted by Prof. Imma Valentina Curato Mar 2025 – May 2025
Chemnitz University of Technology Chemnitz, Germany

Teaching Tutor – *Quantitative Finance and Derivatives* Oct 2024 – Dec 2024
Università degli Studi di Firenze, MSc in Finance and Risk Management Florence, Italy

Professional Experience

Intern – Clinical Risk Management and Patient Safety Unit 2024 – 2026
Azienda USL Toscana Nord Ovest Tuscany, Italy
Collected and analysed patient-safety data and carried out the related statistical analyses. Took part in the unit's meetings to gain first-hand insight into clinical risk management and patient-safety processes.

Conferences, Workshops and Schools

- DOCRA – Doctoral Colloquium on Risk Analytics, AI for Risk — Venice, Sep 2025
- Workshop on Non-parametric Estimation of Stochastic Volatility Models and Climate Econometrics — Florence, Oct 2024
- Workshop on Emerging Risks in Economics and Finance — Florence, Sep 2024
- 42nd Finnish Summer School on Probability and Statistics — Lammi, Finland, May 2024
- XXV Workshop on Quantitative Finance — Bologna, Apr 2024

Doctoral Courses and Seminars

- Bibliometric Analysis and Literature Reviews: Theory and Practice — Lucca, 2024
- Experimental Design and Analysis — Florence, 2024
- La terza missione: dalla ricerca all'impatto — Florence, 2024
- Privatization and Liberalization of Public Utilities — Florence, 2024
- Donation and Prosocial Behaviors — Florence, 2024
- SMEs and Start-ups Engagement with Sustainability — Florence, 2024

Skills

Programming	Python (NumPy, pandas, scikit-learn, XGBoost, Optuna), MATLAB
Finance	Quantitative finance, risk management, option pricing and derivatives valuation, stochastic calculus, risk measures (VaR, ES)
Statistics & ML	Survival analysis, GLMs, gradient boosting, time series (ARIMAX), Monte Carlo simulation, path signatures
Tools	LaTeX, Excel

Languages

Italian, Albanian	Native
English	Fluent
French, German	Working knowledge

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